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Project Title: Effects of Coil Pitch and Flow Configuration on Heat Transfer in a Helically Coiled Heat Exchanger

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Abstract:

This study experimentally investigates the effects of coil pitch (19 mm, 24.5 mm and 30 mm), flow configurations (parallel and counter-current flow), water flowrate (1 and 2 L/min), air velocity (10 and 18.5 m/s) and pulsed and steady airflow on the overall heat transfer coefficient (U) of a helically coiled shell-and-coil heat exchanger. Counter-current flow consistently outperformed parallel flow by an average of 14.7% under steady conditions and was therefore chosen for all analysis. Air velocity was the dominant variable, raising U by circa 60% when increased from 10 to 18.5 m/s. Increasing water flowrate from 1 to 2 L/min generally reduced U , as the resulting drop in LMTD outweighed the gain in waterside convection. The highest measured U was 0.245 kW/m²K at 24.5 mm pitch under steady, counter-current flow at 18.5 m/s and 2 L/min. Each coil pitch behaved as a distinct heat exchanger whose relative performance depended on operating conditions – no single pitch was universally optimal. Pulsed airflow at $St = 0.2$ generally reduced U – only two heat exchanger combination showed enhancement. Theoretical optimisation using Nusselt number correlations identified 19 mm pitch at minimum air velocity and maximum water flowrate as the highest theoretical U of 0.305 kW/m²K, diverging from the experimentally measured peak at 24.5 mm pitch, showing limitations in the airside correlations missing the inter-coil spacing effects on airside convection.

Nomenclature

Symbol	Unit	Description
Geometry		
A	m ²	Area
d	m	Tube or Shell Diameter
D	m	Coil Diameter
H	m	Axial Coil length
L	m	Length of Tube
P	m	Coil Pitch
N	-	Number of turns in coil
t_w	m	Tube Wall Thickness
Heat Transfer		
Q	W	Heat Transfer Rate (Duty)
U	kW/m ² K	Overall Heat Transfer Coefficient
α	kW/m ² K	Specific Heat Transfer Coefficient
k	kW/mK	Thermal Conductivity
LMTD	K	Log Mean Temperature Difference
C_p	kJ/kgK	Specific Heat Capacity
$\dot{m}$	kg/s	Mass Flowrate
$V(u)$	m/s	Fluid Velocity (St Correlation)
ρ	kg/m ³	Fluid Density
μ	kg/m. s	Dynamic Viscosity
f	Hz	Frequency
T	K	Temperature

Symbol	Description
Dimensionless Numbers	
Re	Reynolds Number
Pr	Prandtl Number
Nu	Nusselt Number
De	Dean Number
St	Strouhal Number
ΔX	Change in X
δ	Curvature Ratio
γ	Pitch Ratio (P/d)
Subscripts	
i	Internal
o	Outer
w	Water Side
Air	Air Side
In	Inlet
Out	Outlet
Min	Minimum
Max	Maximum

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1. Introduction

1.1. Context and Background

Heat exchangers are designed to transfer thermal energy between two fluids without allowing them to mix. In this project, this principle is applied using a helically coiled tube, where hot water flows internally and ambient air is externally forced over the coil. In a typical example such as an air conditioning unit, room-temperature air flows over tubes carrying a colder fluid, transferring heat due to the temperature difference between them without direct fluid contact.

The use of a helical coil rather than a straight tube or tubes allows for a more compact heat exchanger design; however, it also significantly changes the flow behaviour within the tube. As the fluid moves through the curved path, centrifugal forces induce a secondary flow motion, described as a “circulatory motion” [1]. This secondary flow improves mixing within the fluid and increases interaction with the tube wall, enhancing the systems overall heat transfer rate.

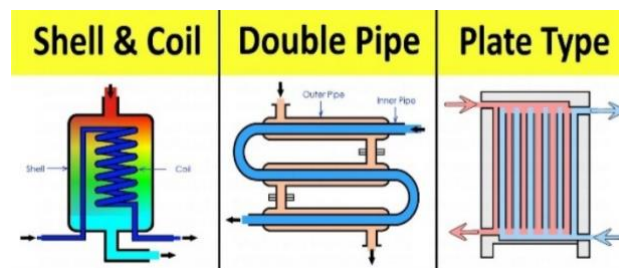


Figure 1- Heat Exchanger Type Comparison [2]

The key geometric parameters of the coil include the pitch (the distance between each turn), the coil diameter and the tube diameter. These parameters directly influence the strength of the secondary flow and therefore the effectiveness of heat transfer of the system. Previous work has shown that optimising these variables can increase the heat transfer coefficient of the water – relative to straight tube designs – by approximately 4 to 14% [3].

1.2. Problem Definition and Knowledge Gap

Although many studies have examined helically coiled heat exchangers, several limitations remain within the existing literature. Most research focuses on internal tube-side behaviour under steady and fully developed flow conditions, with significantly less attention given to shell-and-coil configurations and the role of coil pitch. As stated by Salimpour, “there are few investigations on the heat transfer coefficients of this kind of heat exchanger considering the geometrical effects like coil pitch” [4], confirming that the effect of pitch in these systems is not fully understood or experimentally validated.

Furthermore, research on pulsating flow has mainly considered internal pipe flow rather than pulsed airflow over the coil. Studies have shown that oscillating flow can significantly influence heat transfer, with reported increases in Nusselt number from “5.97 to 28.45” [5], highlighting the potential for strong heat transfer enhancement – the Nusselt number being the “ratio of convective to conductive heat transfer in a fluid” [6]. Previous studies are primarily limited to internal flow systems and do not consider external airflow over a helically coiled geometry.

The combined effect of coil pitch, flow configuration (parallel vs counter-current), and unsteady external airflow has therefore not been sufficiently investigated. This gap in understanding motivated the experimental work carried out in this study.

1.3. Project Aim and Objectives

The aim of this project is to determine how coil pitch and flow configuration influence the heat transfer performance of a helically coiled shell-and-coil heat exchanger, using hot water flowing through the coil and ambient air forced over it. The primary objectives are:

- Design and manufacture snap-fit inserts to fix the coil pitch at 19 mm, 24.5 mm and 30 mm, while ensuring the coil remains central within the shell to maintain consistent airflow distribution.
- Carry out experiments measuring water inlet and outlet temperatures, along with air inlet (ambient) and outlet temperatures for each pitch, water, and air flowrates and flow configuration for both steady and pulsed airflow.
- Quantify the performance difference between parallel and counter-current flow configurations and identify the most effective arrangement.
- Select and apply the appropriate Nusselt number correlations for both water and air sides of the tube to support the analysis of the experimental data.
- Evaluate the effect of pulsed airflow on the heat transfer performance relative to steady airflow conditions.
- Identify the operating and geometric conditions that maximise the overall heat transfer coefficient.

1.4. Significance and Sustainability of Helically Coiled Heat Exchangers

Helical coil heat exchangers are used in batch heating, evaporation, condensation and heat recovery applications, where compactness and thermal efficiency are the primary design considerations. By identifying the optimal pitch and flow configuration for given operating constraints, this study supports the design of more efficient compact heat exchangers.

Improving heat exchanger performance reduces operational energy demand and associated emissions. The rig also minimised material waste by incorporating environmental sustainability: a recycled aluminium shell casing, a repurposed copper coil, and 3D-printed recyclable PLA snap inserts.

2. Literature Review

2.1. Heat Transfer in Helically Coiled Tubes

In a straight tube heat exchanger, the overall heat transfer coefficient is primarily dependent on flow velocity and fluid properties. On the contrary, helically coiled tubes introduce centrifugal forces due to their curved geometry, which generates secondary flow – vortices rotating within the tube cross section – enhancing mixing and overall heat transfer performance [7].

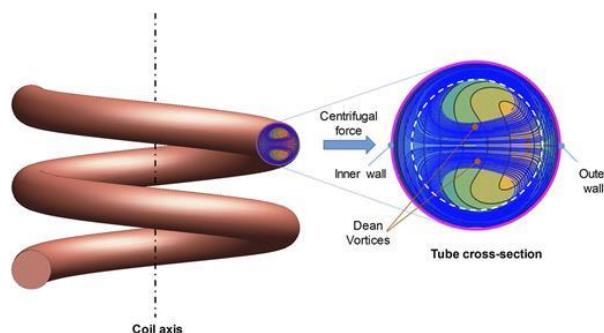


Figure 2 - Secondary Flow Visualisation [8]

As fluid flows through the curved tube, the velocity distribution across the cross-section is altered due to the “circulatory motion” [1], which disrupts the thermal boundary layer and increases convective heat transfer. The curved geometry also increases the effective flow path length and strengthens fluid mixing, further contributing to improved thermal performance. Research shows that the curvature ratio, defined by the relationship between the tube diameter (d) and coil diameter (D), strongly affects the intensity of the secondary flow [9]. This secondary flow is represented using the Dean number, which combines the effects of Reynolds number and curvature ratio.

Performance is assessed using three key dimensionless parameters: the Dean number (De) represents the influence of curvature and flow velocity on secondary flow strength; the Nusselt number (Nu) relates convective heat transfer to conductive heat transfer within the fluid; the Reynolds number (Re) determines whether the flow is laminar, transitional or turbulent. Dimensionless number equations and significance are shown in Table 1. With respect to straight tubes, helically coiled tubes provide a delayed transition to turbulent flow, suggesting that heat transfer can be sustained at higher flowrates within laminar flow regimes [7].

Dimensionless Number	Equation	Significance
Dean Number	$De = Re\sqrt{\delta}$	Secondary flow intensity
Nusselt Number	See correlations in section 2.6.2.	Heat transfer between surface and fluid
Reynolds Number	$Re = \frac{(\rho V)_{fluid} d_{(shell\ or\ coil)}}{\mu_{fluid}}$	Fluid flow state
Prandtl Number	Interpolated from fluid property tables	Related to internal fluid momentum

Table 1 - Dimensionless Number Expansion

2.2. Effects of Coil Pitch and Geometry

The geometry of a helically coiled heat exchanger, particularly coil pitch, strongly influences heat transfer performance where “increasing coil pitch decreases the overall heat transfer coefficient” [10]. Key geometric parameters such as tube diameter, coil diameter, and coil pitch appear directly in dimensionless group correlations, including the curvature ratio ($\delta=d/D$) and pitch ratio (P/d), which govern both internal flow behaviour and external flow interaction [11].

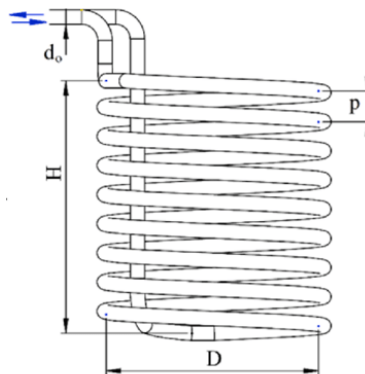


Figure 3 - Coil Geometry [12]

Reducing coil pitch creates a more compact geometry, increasing the number of turns per unit length and intensifying secondary flow effects, which increases the heat transfer coefficient [13]. However, a tighter

pitch also increases internal flow resistance due to stronger curvature effects and reduced spacing between coil turns, leading to a larger pressure loss [7]. In addition, reducing the spacing between turns can restrict external airflow access to the coil surface, which may reduce airside convection. On the contrary, increasing the coils pitch reduces the heat transfer per unit volume and reduces the secondary flow intensity, potentially reducing thermal performance. The optimal geometry therefore balances the enhanced heat transfer and hydraulic performance [14].

In practical systems, maintaining a consistent pitch is challenging due to the vibration from the working fluids, thermal expansion, and manufacturing tolerances. Irregular spacing between coil turns can introduce non-uniform flow distribution, degrading overall performance. Salimpour [15] confirmed that pitch influence remains relatively unexplored in shell-and-tube configurations.

2.3. Flow Configurations in Heat Exchangers

Flow configuration, whether the hot and cold fluids travel in the same (parallel) or opposite (counter-current) directions, significantly affects the temperature gradient available to drive the heat transfer. In parallel flow both fluids enter at the same end, so the temperature difference progressively decreases along the coil, whereas counter-current flow maintains a greater and more uniform temperature difference along the heat exchanger length, raising the LMTD and therefore the overall heat transfer rate. Studies have confirmed that counter-current arrangements consistently produce a higher U under equivalent conditions [16], which, after experimental confirmation, was chosen for all subsequent analysis.

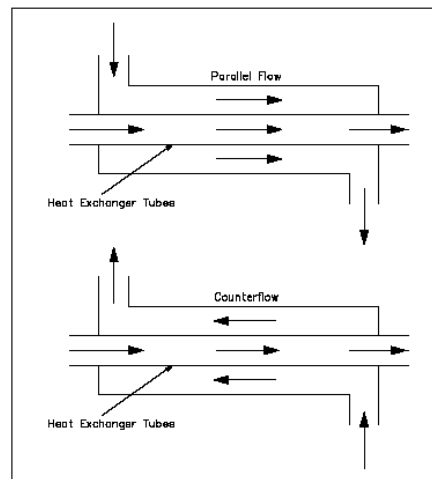


Figure 4 – Parallel and Counter-current Flow Comparison [17]

2.4. Pulsed and Steady Flow

In conventional heat exchangers, flow is steady, producing stable temperature gradients and heat transfer rates. Introducing periodic disruption to the flow can alter boundary layer development and modify the heat transfer behaviour. Published literature shows that pulsating flow can significantly increase the Nusselt number – with increases shown to be from “5.97 to 28.45” [5], representing an almost fivefold enhancement. However, the effect is highly dependent on oscillation frequency, amplitude, and flow regime, where the optimal frequency is determined by the Strouhal number ($St = fd/u$) [5]. Additionally, published studies primarily consider internal pulsing flow, and experimental investigation of pulsed external flow over helically coiled shell-and-tube geometry is substantially limited.

2.5. Material and Design Considerations

Copper was selected for the helical coil due to its relatively high thermal conductivity (0.385 kW/mK) [18] and ductility, which allows the coil’s geometry to be formed and adjusted without material failure. These

properties make it suitable for this investigation, where the pitch is repeatedly changed between test conditions.

The shell housing was a repurposed steel casing, modified with internal insulation to fabricate a 155mm x 155mm internal cross section shell for the heat exchanger. Gaps were sealed off to minimise heat loss to the surroundings and maintain a controlled airflow across the cross-section over the coil.

Maintaining a consistent and reproducible coil pitch was a central design challenge. To address this, snap-fit 3D-printed inserts were designed to lock the coil into three predetermined pitch configurations (19mm, 24.5mm & 30mm), while also centralising the coil axially within the shell housing to ensure uniform airflow distribution.

2.6. Correlations

2.6.1. Thermal correlations

To analyse the experimental data, correlations were required for the main thermal calculations and for the estimation of convective heat transfer coefficients on both the water and air sides. These correlations were used to calculate the heat transfer duty, the log mean temperature difference, and the overall heat transfer coefficient.

To calculate the temperature of the air outlet:

$$T_{air,out} = \frac{(\dot{m}c_p\Delta T)_w}{(\dot{m}c_p)_{air}} + T_{air,in}$$

Equation 1 - Air Outlet Equation

LMTD of a counter-current flow and parallel flow system:

$$\Delta T_{m,cc} = \frac{(T_{water,out} - T_{air,in}) - (T_{water,in} - T_{air,out})}{\ln\left(\frac{T_{water,out} - T_{air,in}}{T_{water,in} - T_{air,out}}\right)} \quad \Delta T_{m,p} = \frac{(T_{water,in} - T_{air,in}) - (T_{water,out} - T_{air,out})}{\ln\left(\frac{T_{water,in} - T_{air,in}}{T_{water,out} - T_{air,out}}\right)}$$

Equations 2, a and b - Counter-current and Parallel flow LMTD

Heat balance of a system:

$$Q = (\dot{m}c_p\Delta T)_{air} = (\dot{m}c_p\Delta T)_w$$

Equation 3 - System Duty and Heat Balance

Overall heat transfer coefficient:

$$U = \frac{Q}{A_{coil}\Delta T_m}$$

Equation 4 - Overall Heat Transfer Coefficient of System

In addition to these system performance equations, Nusselt number correlations were considered for both water and airside to estimate the individual convective heat transfer coefficients. Since several published correlations are available for helically coiled heat exchangers, the most appropriate equations were selected by comparing their theoretical results against the experimental overall heat transfer coefficient. The selected correlations are later used in the methodology and results sections to calculate the water and air side heat transfer coefficients.

2.6.2. Nusselt number correlations

The following correlations (Table 2) were identified from the literature for internal flow in helical coils and considered for this study.

Correlation Number	Reference	Equation	Limits	Comments
$Nu_{w,theo}$ (Equation 5)	Dittus-Boelter	$Nu = 0.023Re^{0.8}Pr^{0.3}$	N/A	Theoretical Nu
$Nu_{w,1}$ (Equation 6)	M.R. Salimpour	$Nu = 0.152De^{0.431}Pr^{1.06}\gamma^{-0.277}$	$De < 3000$	Developed for shell and tube heat exchangers
$Nu_{w,2}$ (Equation 7)	Kalb et al	$Nu = 0.836De^{0.5}Pr^{0.1}$	$De \geq 80$; $0.7 < Pr < 5$	Laminar flow in tight tubes
$Nu_{w,3}$ (Equation 8)	Xin et al	$Nu = (2.153 + 0.318De^{0.643})Pr^{0.177}$	$20 < De < 2000$; $0.7 < Pr < 175$; $\delta < 0.0884$	Accounts for curvature effect
$Nu_{w,4}$ (Equation 9)	Roger et al	$Nu = 0.023Re^{0.85}Pr^{0.4}\delta^{0.1}$	$Re > 2000$	General turbulent flow

Table 2 – Waterside Nusselt Number Correlations – Denoted as $Nu_{w,x}$

For external flow over the helical coil within the shell, the following correlations (Table 3) were considered.

Correlation Number	Reference	Equation	Comments
$Nu_{air,1}$ (Equation 10)	M.R. Salimpour	$Nu = 19.64Re^{0.513}Pr^{0.192}\frac{D}{d}^{-0.938}$	Developed for shell-side Nusselt Number
$Nu_{air,2}$ (Equation 11)	Moawed	$Nu = 67.087Re^{0.197}Pr^{0.128}\frac{P}{d}^{-0.528}$	Developed for shell-side Nusselt Number
$Nu_{air,3}$ (Equation 12)	Alimoradi and Veysi	$Nu = 0.45Re^{0.48}Pr^{1.06}\frac{D}{d}^{0.914}\frac{P}{d}^{0.281}$	Developed for shell-side Nusselt Number

Table 3 – Airside Nusselt Number Correlations – Denoted as $Nu_{air,x}$

The correlations to calculate the theoretical heat transfer coefficients of air and water is as follows:

$$\alpha_w = \frac{Nu k_w}{d_{coil}} \quad \alpha_{air} = \frac{Nu k_{air}}{d_{shell}}$$

Equation 13; a & b - Fluid Heat Transfer Coefficient

To confirm the most accurate Nusselt number correlations gathered from literature, the equation is used to calculate theoretical overall heat transfer coefficient, using every combination of each Nusselt number equations for water and air. Using the waterside Nusselt number correlations to calculate α_{water} and the airside correlations to calculate α_{air} , the $U_{theoretical}$ equations is as shown below:

$$U_{Theoretical} = \frac{1}{\frac{1}{\alpha_w} + \frac{1}{\alpha_{air}}}$$

Equation 14 - Theoretical Overall Heat Transfer Coefficient

The simplified thermal resistance overall heat transfer coefficient correlation in Equation 14 was applied opposed to a more sophisticated correlation including the geometry and material properties of the coil.

However, as the coil satisfies the thin-walled correlation ($t_w < 0.1d_o$) and has high conductivity properties, changing the resistance correlation would provide a negligible improvement in accuracy, while adding unnecessary uncertainty in assuming a geometrically perfect coil.

The theoretical U is compared against the experimental U to comprehend the most accurate pair of correlations from literature. An example table for the approach of this is as seen below in Table 4:

		$X_{pitch}, Y_{Flowrate}, Z_{velocity}$				
		α_{water}				
	$U_{exp} = X$	$Nu_{w, theo}$	$Nu_{w,1}$	$Nu_{w,2}$	$Nu_{w,3}$	$Nu_{w,4}$
α_{air}	$Nu_{air,1}$	$U_{Theo, 1}$	$Nu_{1,1}$	$Nu_{2,1}$	$Nu_{3,1}$	$Nu_{4,1}$
	$Nu_{air,2}$	$U_{Theo, 2}$	$Nu_{1,2}$	$Nu_{2,2}$	$Nu_{3,2}$	$Nu_{4,2}$
	$Nu_{air,3}$	$U_{Theo, 3}$	$Nu_{1,3}$	$Nu_{2,3}$	$Nu_{3,3}$	$Nu_{4,3}$
% difference						
α_{air}	$Nu_{air,1}$	%	%	%	%	%
	$Nu_{air,2}$	%	%	%	%	%
	$Nu_{air,3}$	%	%	%	%	%

Table 4 – Nusselt Number Accuracy Comparison Example

Where the theoretical U values containing the least divergence from the experimental value for both water and air is used to process the heat transfer coefficients for both fluids in the results section.

3. Methodology

3.1. Experimental Rig Design

The experimental rig consists of a copper helical coil mounted inside a rectangular steel housing. The housing is fitted with insulation and aluminium tape to create a 155mm x 155mm shell and eliminate gaps, in turn preventing airflow leakage and ensuring a controlled flow path over the coil. The coil is centrally positioned within the shell via the 3D-printed snap fit inserts, which both fix the coil pitch and maintain a consistent axial position to ensure uniform airflow distribution. The housing acts as the shell of the heat exchanger through which air is forced by a centrifugal fan at one end. The coil's water inlet and outlet connections are both located at the air-inlet side of the shell and are swapped to vary the configuration between counter-current and parallel flow.

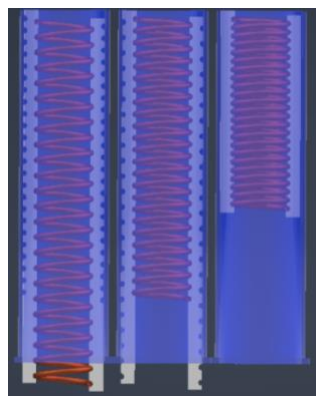


Figure 5 - Coil Pitch Comparison

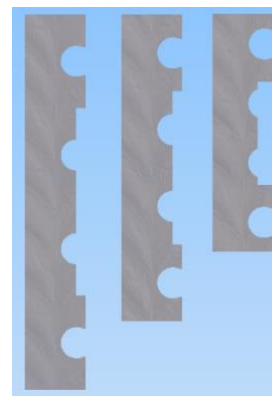


Figure 6 - Final Pitch Insert Iteration

The coil was wound from a 10 mm outer diameter, 9 mm inner diameter, copper tube with a coil diameter of 135 mm. At 19 mm pitch the axial length is 437 mm, at 30 mm pitch it is increased to 690 mm. The total

tube length is 9.77 m [19], giving a consistent external heat transfer area of $A_{coil} = \pi d_o L$, resulting in a surface area of 0.307 m² across all pitches. Where L is defined as $L = \pi D_{coil} N$, giving a coil length of 9.77 m. Hot water enters the coil at approximately 60°C supplied from a mains and boiler water source. Ambient air is forced through the housing via a centrifugal fan. For counter-current tests, the water inlet is reversed relative to the fixed air inlet, whereas for parallel flow both fluids enter and flow in the same direction through the shell.

3.2. Geometric Correlations

Curvature ratio:

$$\delta = d/D = \frac{0.01}{0.135} = 0.0741$$

Equation 15 - Curvature Ratio

Internal tube cross sectional area:

$$A_{CSA,coil} = \pi r_i^2$$

Equation 16 - Coil Cross Sectional Area

External surface area:

$$A_{surf} = \pi d_o L$$

Equation 17 - Coil External Surface Area

3.3. Instrumentation and Data Acquisition

Two K-type thermocouples were positioned at the water inlet and the water outlet. Ambient air temperature was measured utilising a built-in air temperature sensor from the mains hot water pump unit. The thermocouples were calibrated by comparing the readings against the ambient temperature sensor to ensure accuracy.

A turbine flowmeter was used to measure the water flowrate, which was set to either 1 L/min or 2 L/min using a manual needle valve. The air velocity through the shell was measured using a manometer at the shell outlet and adjusted by restricting the airflow using a plate at the vent outlet. A fully open vent produced a maximum velocity of 18.5 m/s, while partial restriction was used to achieve 10 m/s. Under pulsed conditions, the airflow was not constant and the anemometer reading fluctuated, therefore the air velocity was taken as the average value over the test period rather than an instantaneous reading.

Temperature readings were recorded at steady state, defined as a temperature deviation of less than 0.5°C over a two-minute period. A minimum of two repeat readings were taken for each condition and averaged to reduce measurement uncertainty. Inlet water and air temperatures varied due to ambient conditions and heat fluctuations (typically 58-63°C for water and 23.3-24.6°C for air), therefore, averaging was required to improve reliability. All temperature data was recorded and exported as a CSV file for processing in a spreadsheet.

3.4. Operating Conditions

Tests were conducted for three coil pitches (19mm, 24.5mm and 30mm) to represent a compressed, un-sprung and decompressed coil, two air velocities (10 m/s and 18.5 m/s), two water flowrates (1 L/min and 2 L/min), and both counter-current and parallel flow configurations. This produced a test matrix of 24 test conditions for each airflow type (steady and pulsed), resulting in 48 total configurations analysed after focusing on the most effective flow arrangement.

Fluid properties for air and water were taken at a bulk temperature value and interpolated from the fluid properties booklet [20] to ensure consistency across all calculations.

3.5. Pulsed Airflow Generation

Pulsed airflow was generated using an electronic timer (Micon 175 series) connected between the mains supply and the fan power cable, which periodically interrupted the fan operation at a fixed frequency. Due to the inertia of the centrifugal fan, the interruption produced a sinusoidal variation in air velocity rather than an ideal square-wave stop-start pattern. The oscillation frequency was determined using the Strouhal number with $St = 0.2$, producing a frequency of 3.3 Hz (a period of 0.3 seconds). Each pulsed test was conducted under the same mean air velocity and water flowrate as its corresponding steady airflow test, enabling direct comparison. The pulsed airflow enhancement ratio is defined as U_{Pulsed}/U_{Steady} where a value greater than 1 indicates that pulsing improved the heat transfer performance.

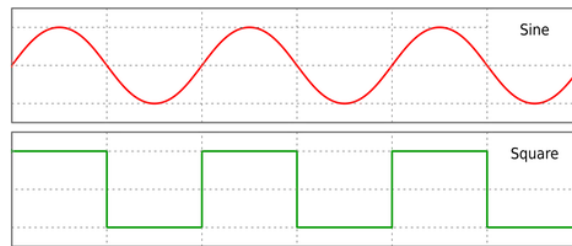


Figure 7 - Resulting Sinusoidal vs Ideal Square Waves [21]

3.6. Assumptions and Limitations

All fluid properties were evaluated at a bulk temperature. Heat losses through the shell walls were assumed negligible due to insulation lining. Airflow was assumed uniform across the shell cross-section. The water flow was assumed developed after the first three complete coil turns. The supplied mains' water temperature varied slightly between tests – this was accounted for by averaging results over multiple tests and represents a minor experimental variable. Oscillation frequency was consistent but not precise due to an analogue dial where the frequency used should be assumed approximate opposed to exact. The table of constant values is as seen below:

Property	Air	Water
Bulk Temperature (°C)	24.1	60.2
Density (kg/m ³)	1.19	983.1
C _p (kJ/kgK)	1.005	4.185
Dynamic Viscosity (kg/m. s)	1.83x10 ⁻⁵	4.65x10 ⁻⁴
Thermal Conductivity (kW/mK)	2.67x10 ⁻⁵	6.53x10 ⁻⁴

Table 5 – Table of Constant Fluid Values

3.7. Health and Safety

Identified hazards included: handling hot water (burn and scald risk), electrical equipment operating within vicinity of water (electric shock risk) and pressure changes in the pipe (spillage and scald risk). Mitigation measures included: use of insulated gloves when connecting/disconnecting waterlines, isolation of electrical connections from wet surfaces, testing connections prior to experimental runs to check for leaks and structurally securing the rig to prevent unwanted displacement. A formal risk assessment was completed and is presented in the appendix.

4. Results

4.1. Data Quality

Steady-state conditions were achieved for each test, with repeated temperature readings varying with $\pm 0.5^\circ\text{C}$, confirming stabilisation before recording – shown in Figure 8. Inlet water temperature remained consistent across all tests (58°C – 63°C); minor variations were accounted for by averaging repeated results. An energy balance comparing waterside and airside heat transfer rates showed reasonable alignment for all cases, given sensor accuracy and interpolated fluid properties.

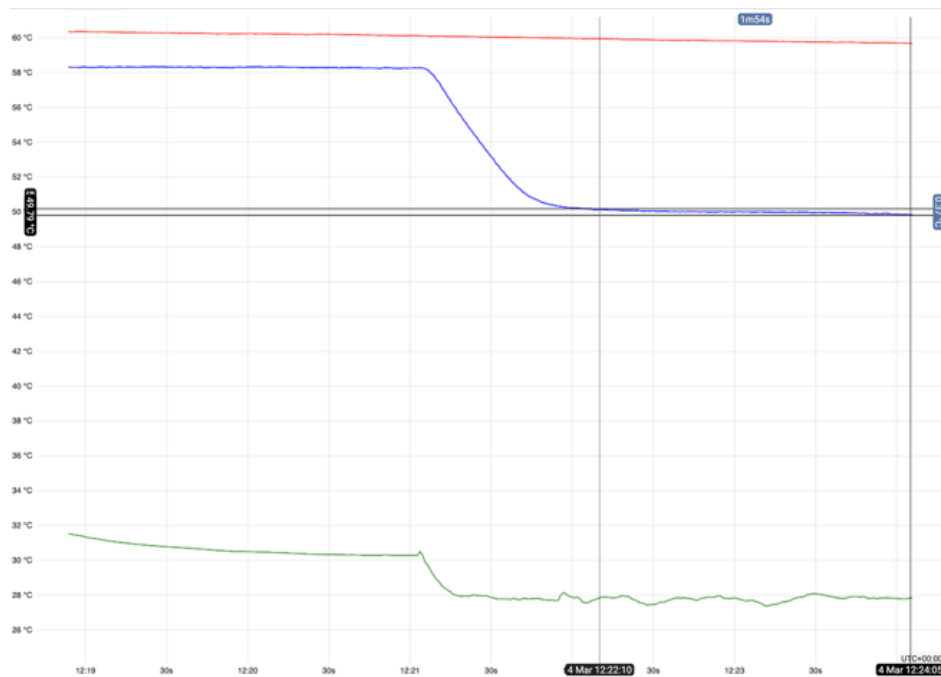


Figure 8 – Pico Log Steady-State Visualisation with a Change in Temperature of 0.377°C

4.2. Parallel vs Counter-Current Flow

Graphs 1 and 2 compare U for parallel and counter-current flow under steady and pulsed airflow, respectively, across all pitch, water flowrate and air velocity combinations. Under steady conditions (Graph 1), counter-current flow produced a higher U in every case, with an average improvement of 14.7%. The largest gain occurring at 30 mm pitch, 1 L/min and 10 m/s (30.2%). Under pulsed conditions (Graph 2), counter-current flow retained its advantage with a reduced improvement of 1% on average. This indicates that flow oscillation can disrupt the shell side temperature distribution significantly to negate the LMTD advantage of counter-current arrangements. Counter-current flow is confirmed as the more effective configuration under steady conditions and is used for this study's analysis.

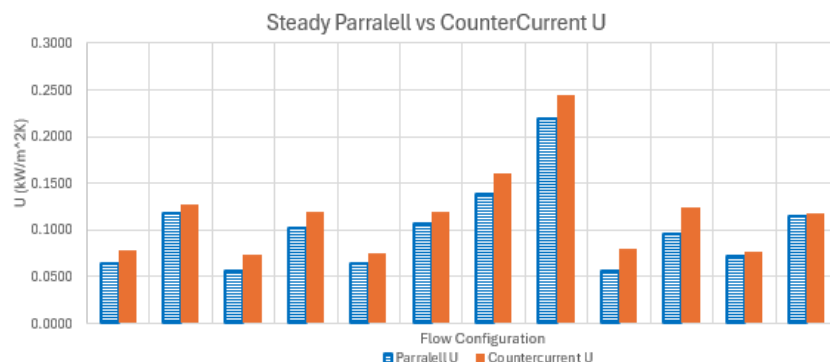


Figure 9 – Graph 1 - Comparing Counter-Current and Parallel Flow for Steady Airflow

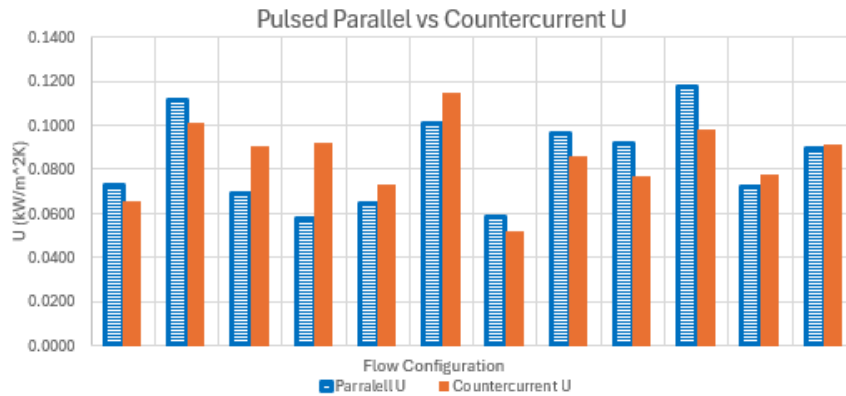


Figure 10 - Graph 2 - Comparing Counter-Current and Parallel Flow for Pulsed Airflow

4.3. Effects of Water Flowrate

Graphs 3-5 show U against water flowrate for the three pitches at each combination of airflow condition and air velocity. For many test conditions, doubling water flowrate from 1 to 2 L/min decreased U rather than increasing it, despite the higher waterside Reynolds number.

At 19 mm pitch, counter-current flow, steady airflow and 10 m/s air velocity – U decreased from 0.0788 kW/m²K at 1 L/min to 0.0741 kW/m²K at 2 L/min: a 6% reduction in U . At 18.5 m/s under steady airflow, U fell from 0.127 kW/m²K at 1 L/min to 0.119 kW/m²K: a 6.8% reduction. This reduction in U – despite a higher Reynolds number – is due to the reduced residence time at 2 L/min – the water temperature drop decreases more than the gain in waterside convection can compensate, reducing Q , and therefore reducing U .

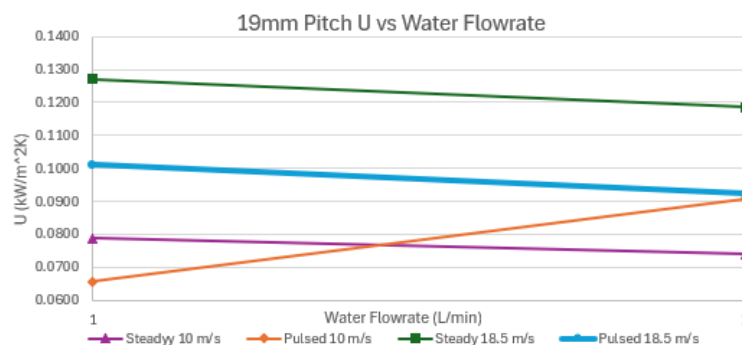


Figure 11 - Graph 3 - 19 mm Pitch: U vs Flowrate

At 24.5 mm pitch, 10 m/s and steady airflow conditions, U increased from 0.075 kW/m²K at 1 L/min to 0.160 kW/m²K at 2 L/min: a 114% increase. A similar increase is observed with the high air velocity of the same conditions. These two results are an apparent anomaly with respect to the other two pitches and may reflect a specific flow interaction between the unsprung pitch geometry and higher water flowrate and air velocity, which may not be apparent in both 19 mm and 30 mm pitches.

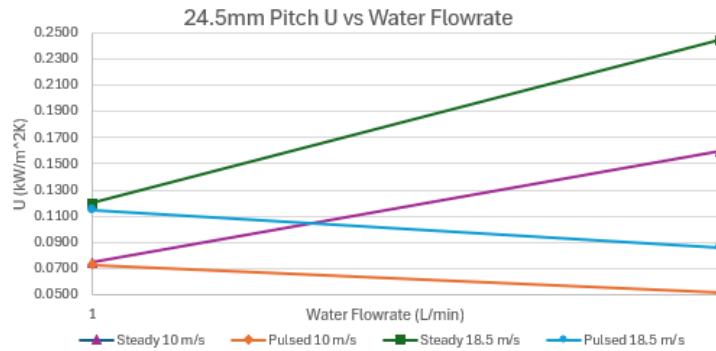


Figure 12 - Graph 4 - 24.5 mm Pitch: U vs Flowrate

At 30 mm pitch, the trend is more consistent where U remains relatively stable between flowrates at 10 m/s for both steady and pulsed airflow. Where for the steady airflow, 10 m/s, U decreased from 0.081 kW/m²K at 1 L/min to 0.076 kW/m²K at 2 L/min: a decrease of 5.6%. At 18.5 m/s under steady airflow, U dropped from 0.124 kW/m²K at 1 L/min to 0.118 kW/m²K at 2 L/min: a decrease of 5%.

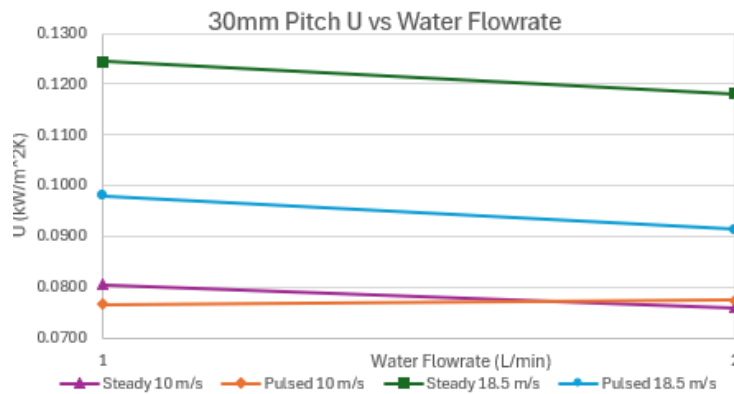


Figure 13 - Graph 5 - 30 mm Pitch: U vs Flowrate

4.4. Effects of Air Velocity

Graphs 6-8 show U against air velocity for the three coil pitches at both steady and pulsed airflow and both water flowrates. Increasing air velocity from 10 m/s to 18.5 m/s produced a substantial increase in U across all conditions. Under steady airflow conditions at 19 mm pitch and 1 L/min, U increased from 0.079 kW/m²K at 10 m/s to 0.127 kW/m²K at 18.5 m/s: a 61.4% increase. Also, at 2 L/min, the corresponding increase was from 0.074 kW/m²K to 0.119 kW/m²K: a 60.1% increase. Under pulsed conditions at 1 L/min, U increased from 0.066 kW/m²K at 10 m/s to 0.101 kW/m²K at 18.5 m/s: a 52.9% increase. Under pulsed conditions at 2 L/min, the change in U across air velocities is negligible and is treated as an anomaly in the results.

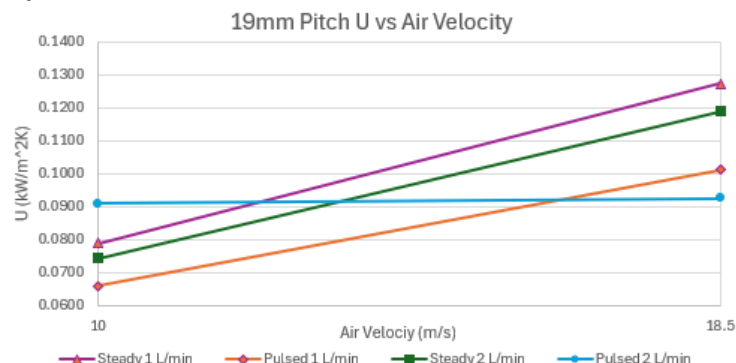


Figure 14 - Graph 6 - 19 mm Pitch: U vs Air Velocity

At 24.5 mm and 30 mm pitches, Graphs 7 and 8, similar positive gradients are observed for all combinations of airflow condition and water flowrate, confirming that increasing air velocity consistently enhances U regardless of pitch. The largest U values across the range of combinations are consistently obtained at 18.5 m/s air velocity. These results confirm that the airside resistance represents the dominant factor of the total thermal resistance in this heat exchanger, and that increasing the airside heat transfer coefficient is the most effective way to improve overall performance.

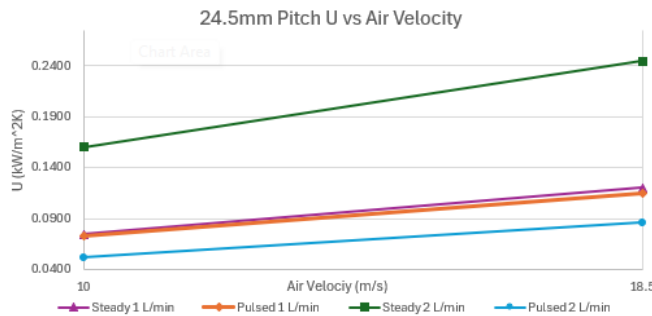


Figure 15 - Graph 7 - 24.5 mm Pitch: U vs Air Velocity

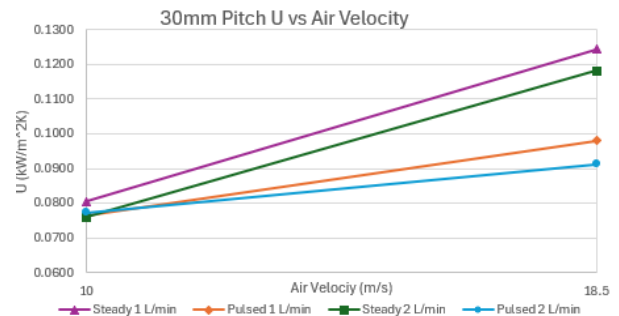


Figure 16 - Graph 8 - 30 mm Pitch: U vs Air Velocity

4.5. Effects of Coil Pitch

Graphs 9 and 10 compare U across the three pitches for all combinations of water flowrate and air velocity under steady and pulsed airflow, respectively.

Under steady airflow (Graph 9), the relationship between pitch and U is non-monotonic and dependent on the operating conditions. At low air velocity and water flowrate, U is similar across all three pitches (0.079, 0.075 and 0.08 kW/m²K for 19 mm, 24.5 mm and 30 mm, respectively), showing that pitch has little influence when both driving forces are at low velocity and flowrate. At low velocity and high flowrate, U increases sharply for the 24.5 mm coil (0.16 kW/m²K) relative to 19 mm (0.074 kW/m²K) and decreases back towards the 30 mm coil (0.076 kW/m²K). At high velocity and low flowrate, U is broadly similar for 19 mm and 30 mm (0.127 and 0.124 kW/m²K) but slightly lower at 24.5 mm (0.12 kW/m²K). The most notable pitch effect occurs at high velocity and high flowrate: U rises from 0.119 kW/m²K at 19 mm to peak at 0.245 kW/m²K at 24.5 mm, before dipping back at 0.118 kW/m²K at 30 mm pitch. This indicates that for the highest air velocity and water flowrate conditions, the intermediate pitch produces substantially higher thermal performance.

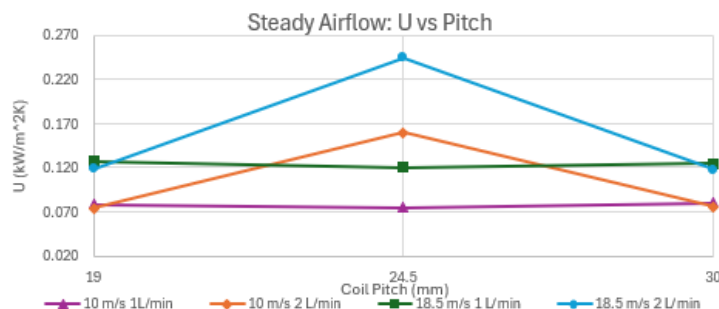


Figure 17 - Graph 9 - Steady Airflow: U vs Pitch

Under pulsed airflow (Graph 10), the trends are less pronounced. At high velocity and low flowrate, U increases from 0.101 kW/m²K at 19 mm to 0.115 kW/m²K at 24.5 mm before declining to 0.098 kW/m²K at 30 mm. At low velocity and high flowrate, U dips sharply at 24.5 mm (0.052 kW/m²K) compared with 19 mm (0.091 kW/m²K) and 30 mm (0.077 kW/m²K). The strong 24.5 mm peak previously observed in the steady, high velocity case is not as clearly represented under pulsed conditions, suggesting that the

unsteady airflow disrupts the pitch dependent flow combinations responsible for the previous peak in performance under steady airflow conditions.

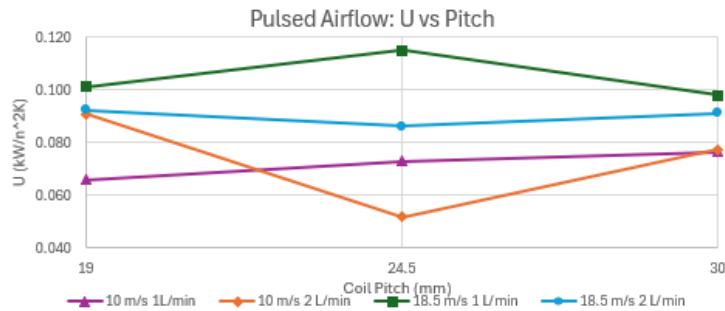


Figure 18 - Graph 10 - Pulsed Airflow: U vs Pitch

Each coil pitch behaves as a distinct heat exchanger of which relative performance depends on the combination of air velocity, water flowrate and airflow conditions. Overall, no single pitch excels over all conditions. To ensure apparent anomalies are not a cause of experimental issues, Graph 11 suggests this isn't the case. Using the most accurate theoretical fluid heat transfer coefficient Nusselt number correlations explained in section 4.6. – intermediate pitches of 22.75 mm and 27.25 mm are chosen (as intermediates between 19 and 24.5 mm, and 24.5 and 30 mm pitch) to calculate the theoretical overall heat transfer coefficient, of which are graphed, confirming that the trends observed in Graphs 9 and 10 are true.

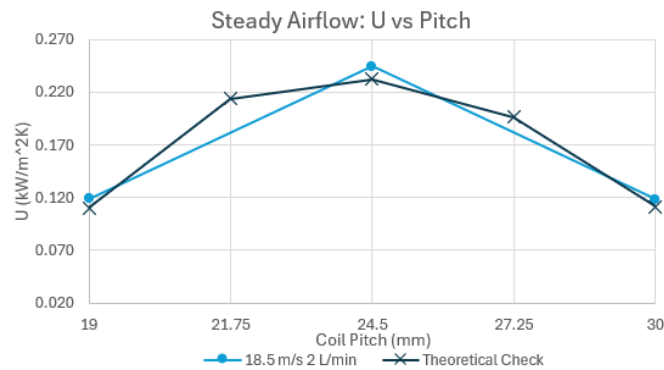


Figure 19 - Graph 11 - Graphical Consistency Check

4.6. Nusselt Number and Fluid Heat Transfer Coefficients

The four waterside and three airside Nusselt number correlations were compared against the experimentally measured overall heat transfer coefficient for all twelve steady airflow test groups. The comparison was carried out by computing the overall heat transfer coefficient by each combination of waterside and airside correlations and converting the result as a percentage of the experimental U value.

		19mm 2 L/min 18.5 m/s Steady Airflow				
		α_{water}				
	U=0.119 kW/m ² K	Nu _{w, theo}	Nu _{w,1}	Nu _{w,2}	Nu _{w,3}	Nu _{w,4}
α_{air}	Nu _{air,1}	0.115	0.111	0.114	0.115	0.115
	Nu _{air,2}	0.116	0.112	0.115	0.116	0.116
	Nu _{air,3}	0.116	0.112	0.115	0.120	0.116
% difference						
α_{air}	Nu _{air,1}	0.967	0.937	0.963	0.972	0.973
	Nu _{air,2}	0.974	0.944	0.971	0.980	0.980
	Nu _{air,3}	0.974	0.944	0.971	1.008	0.980

Table 6 - 19 mm Nusselt Number Correlation Comparison

		24.5mm 2 L/min 18.5 m/s Steady				
		α_{water}				
	U=0.2446 kW/m ² K	Nu _{w, theo}	Nu _{w,1}	Nu _{w,2}	Nu _{w,3}	Nu _{w,4}
α_{air}	Nu _{air,1}	0.228	0.215	0.227	0.224	0.231
	Nu _{air,2}	0.232	0.218	0.230	0.235	0.235
	Nu _{air,3}	0.218	0.230	0.230	0.234	0.235
% difference						
α_{air}	Nu _{air,1}	0.863	0.748	0.850	0.827	0.887
	Nu _{air,2}	0.894	0.776	0.881	0.917	0.919
	Nu _{air,3}	0.773	0.877	0.877	0.913	0.915

Table 7 - 24.5 mm Nusselt Number Correlation Comparison

		30 mm 2 L/min 18.5 m/s Steady				
		α_{water}				
	U=0.118 kW/m ² K	Nu _{w, theo}	Nu _{w,1}	Nu _{w,2}	Nu _{w,3}	Nu _{w,4}
α_{air}	Nu _{air,1}	0.114	0.111	0.114	0.115	0.115
	Nu _{air,2}	0.115	0.112	0.115	0.116	0.116
	Nu _{air,3}	0.115	0.111	0.115	0.116	0.116
% difference						
α_{air}	Nu _{air,1}	0.967	0.937	0.964	0.972	0.973
	Nu _{air,2}	0.975	0.945	0.972	0.980	0.981
	Nu _{air,3}	0.973	0.943	0.970	0.979	0.979

Table 8 - 30 mm Nusselt Number Correlation Comparison

From Tables 6, 7 and 8, the three most effective correlation combinations are Nu_{air,2} with Nu_{w,3} or 4 or Nu_{air,3} with Nu_{w,4}. The most consistently effective correlation combination is Nu_{air,2} with Nu_{w,4}, which is the combination used in this study to calculate the heat transfer coefficient of both water and air.

4.7. Effects of Pulsed Airflow

The impact of pulsed airflow relative to steady airflow is quantified by the enhancement ratio ($U_{\text{Pulsed}}/U_{\text{Steady}}$) and presented in Graphs 12 and 13.

Graph 12 (1 L/min water flowrate) shows that pulsing reduces U in all twelve combinations of pitch and air velocity. At 1 L/min, the enhancement ratio is below 1 in every case, with reductions of 16.5%, 20.5%,

2.3%, 4.1%, 4.9% and 21.2% across the six combinations for 19 mm, 24.5 mm and 30 mm pitches at 10 and 18.5 m/s, respectively. The largest negative effects occur at pitches 19 mm and 30 mm over all air velocities – excluding 30 mm pitch with low air velocity.

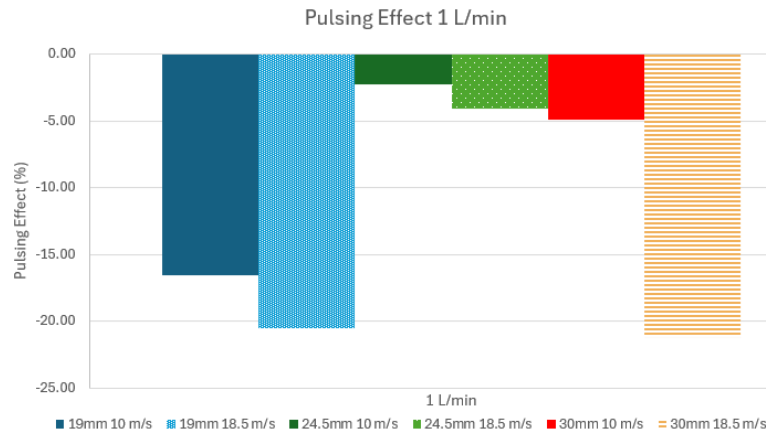


Figure 20 - Graph 12 - Effects of Pulsing at 1 L/min Flowrate

Graph 13 (2 L/min water flowrate) presents a slightly more varied insight. For 19 mm pitch at 10 m/s, the enhancement ratio is 1.224 (+22.4% enhancement), and for 30 mm pitch at 10 m/s the ratio is 1.018 (+1.8%), indicating marginal improvement. For all other combinations of pitch and air velocity at 2 L/min, the enhancement ratio is below 1. Notably, the 24.5 mm pitch produces the largest reductions under pulsed conditions of 67.7% and 64.8% at 10 and 18.5 m/s, respectively. These large decreases at 24.5 mm and 2 L/min consistent with high measured U for this coil under steady conditions, suggesting that oscillation disrupts the specific flow structures responsible for the performance at this pitch.

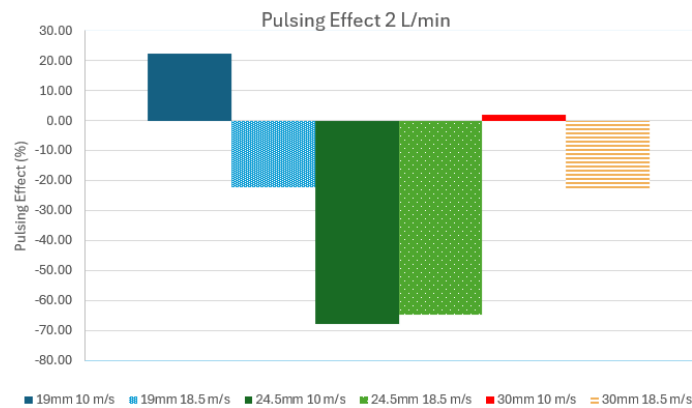


Figure 21 - Graph 13 - Effects of Pulsing at 2 L/min Flowrate

Overall, pulsed airflow at the chosen Strouhal number ($St=0.2$) does not provide a systematic improvement in heat transfer performance for this heat exchanger geometry. Of the twelve pitch and velocity combinations tested at 2 L/min, only two showed a performance enhancement when pulsing, with both gains being small, relative to the reductions observed elsewhere. Reliable pulsed flow enhancement would require careful optimisation of the oscillation frequency and waveform for the specific geometry.

4.8. Optimised Operating Point

An optimisation analysis was performed using the selected Nusselt number correlations to identify the combination of pitch and flowrate conditions that maximises the calculated overall heat transfer coefficient for steady airflow. The objective was to maximise U across the three pitches, utilising the air

velocity and water flowrate constraints set in this report, further using the experimental temperature as constraints. The optimisation was carried out using *Excel Solver*.

The optimised system identified 19 mm pitch at minimum air velocity and maximum water flowrate as producing the highest theoretical U (0.305 kW/m²K), driven by the higher Dean number and stronger secondary flow at the compressed pitch. This diverges from the experimentally measured peak of 0.245 kW/m²K at 24.5 mm – where the airside correlation fails to account for restricted external airflow access at a tighter coil spacing. Both findings are additive: the optimisation identifies the theoretically optimal pitch from the correlations, while the experimental data identifies the pitch that produced the highest measured U, together providing a more complete system design benchmark.

5. Discussion

5.1. Geometric Effects

The experimental results confirm that the three coil pitches behave as three distinct heat exchangers of which relative performance is strongly dependent on the operating point. No single pitch dominated across all conditions, and the dominant geometry varied with air velocity, water flowrate and airflow conditions.

Under steady airflow at 18.5 m/s and 2 L/min – the conditions that produced the highest U values in this study – the 24.5 mm pitch outperformed both the 19 mm and 30 mm pitches, achieving 0.245 kW/m²K. This peak at the intermediate pitch may be due to a balance between two effects: sufficient coil compactness to generate a strong secondary flow, and sufficient spacing between coil turns to allow external flow to access a greater portion of the coil surface area. The 19 mm coil, though geometrically tighter and producing stronger internal secondary flow, appears to restrict external flow access due to the reduced spacing between the turns, reducing the airside heat transfer coefficient. The 30 mm coil provides better external flow access but produced a weaker secondary flow and lower effective surface area per unit length, reducing thermal performance at high flowrates.

At lower air velocity and water flowrate, the pitch effect is less pronounced and the differences between the three coils are small (± 0.006 kW/m²K between the three pitches). Under these conditions, neither internal secondary flow nor external airflow access is a limiting factor, resulting in all three pitches performing similarly to one another. The 30 mm pitch showed some of the largest percentage gains when operating conditions were improved – reflecting its weaker absolute baseline performance, which makes it more sensitive to experimental modifications.

5.2. Flow Configuration and Thermal Performance

The counter-current flow advantage, averaging 14.7% improvement in U under steady conditions, is consistent with the higher LMTD maintained throughout the heat exchanger length. The largest gain was observed for the 30 mm coil at 1 L/min and 10 m/s (30.2%), reflecting that the weakest performing configuration benefits most from an improved thermal driving force. Under pulsed airflow, the average advantage fell to 1% enhancement and, in most cases, counter-current flow produced a lower U than pulsed parallel flow – demonstrating that oscillation can modify the airside temperature distribution sufficiently to negate the LMTD advantage of counter-current configuration.

5.3. Flowrate and Velocity Effects

The difference between the effects of water flowrate and air velocity on U is explained by the dominant resistance in the system. Since $\alpha_{\text{air}} \approx 0.15\text{--}0.23$ kW/m²K is approximately an order of magnitude lower than $\alpha_{\text{w}} \approx 1.3\text{--}4.0$ kW/m²K measured across the range of experiments, the airside thermal resistance accounts for most of the total resistance term (1/U). As a result, increasing the airside convective heat transfer

coefficient by raising air velocity, and therefore Re_{air} , produced a substantial improvement in U , whereas increasing the waterside flowrate provides limited additional benefit.

The reduction in U at higher water flowrate is a consequence of the log mean temperature difference. At 2 L/min, the shorter water residence time in the coil reduces the water temperature drop. This decreases the LMTD more than the gain in waterside heat transfer coefficient can compensate, resulting in a lower Q and therefore a lower U – even though the waterside Reynolds number has increased.

The 114% increase in U at 24.5 mm, 10 m/s between 1 and 2 L/min (0.075 to 0.16 kW/m²K) remains the standout result. The data is consistent and repeatable within the measurement uncertainty, suggesting a real physical effect. One possible explanation is that the interaction between the high flowrate and increased secondary flow combined with the external flow access of an intermediate pitch produced a particularly effective heat transfer configuration. Further investigation with pressure drops measurements would be required to confirm this.

5.4. Pulsed Airflow

The pulsed airflow enhancement ratios show that at the chosen oscillation variable ($St = 0.2$), unsteady external airflow does not systematically improve heat transfer performance. Only two operating points showed enhancement: 19 mm pitch at 10 m/s with 2 L/min (enhancement ratio of 1.224) and 30 mm pitch at 10 m/s with 2 L/min (enhancement ratio of 1.018). These conditions share the feature of a low air velocity combined with a high-water flowrate, suggesting that modest oscillation at low air velocity can improve the airside boundary layer disruption to improve the overall heat transfer coefficient of the system. On the contrary, at 18.5 m/s, where the steady airside heat transfer coefficient is already high, the same oscillation values introduced an enhancement ratio of less than 1.

Since the Nusselt number correlations are evaluated at the same mean velocity for both steady and pulsed tests, the theoretical U does not change significantly between the two conditions. Any measured difference in U reflects an uncaptured boundary layer effect introduced by oscillation – something not shown in existing steady-flow correlations.

The large reductions in U at 24.5 mm pitch under 2 L/min pulsed conditions (circa 65%) indicates that oscillation disrupts the airflow interaction which produced the 24.5 mm peak under steady conditions. The intermediate pitch which is the most sensitive to external airflow conditions under steady flow, appears equally the most vulnerable to performance degradation when the airflow is disturbed.

5.5. Design Implications

The results provide several practical guidelines for the design of compact shell-and-coil heat exchangers utilising forced air over a coil carrying water.

Counter-current flow and high air velocity are the two most reliable methods of maximising U and should be prioritised in any design where high air velocity and a compact tube layout are requirements. Air velocity contributes the largest single improvement in U across all conditions (circa 55%).

Pitch selection should be made with reference to the anticipated operating conditions rather than reducing pitch for maximum compactness. The 24.5 mm pitch delivers the highest measured U under steady high velocity, high flowrate conditions; the 19 mm pitch performs well at high velocity and lower flowrate. Increasing water flowrate is not an effective method in coils with similar geometric characteristics of this rig, as the reduction in LMTD outweighs any gain in internal heat transfer coefficient.

5.6. Limitations

Several limitations should be acknowledged when interpreting these experimental findings. Only three coil pitches were tested, where the intermediate pitch shows superior performance in certain high air velocity conditions, the optimal pitch could lie between the tested values or outside this range. Pressure drop measurements were not recorded, meaning the thermo-hydraulic performance of each configuration cannot be fully checked, as tighter coils and higher water flowrates are likely to induce a higher pumping requirement even when they deliver a higher U .

The pulsed airflow system, based on a mains timer and centrifugal fan, did not deliver the intended stop-and-start airflow pattern. Instead, it delivered a sinusoidal wave pattern due to the inertia of the fan preventing an instantaneous halt in airflow. The effective amplitude and frequency experienced by the coil were therefore approximate, limiting direct comparison with literature that assumes square wave oscillations. All conclusions are based on a bulk ambient, inlet and outlet temperature, which are appropriate for overall energy balance but do not expose local variations in heat transfer along the coil's length. Despite these limitations, the general consistency of the trends and the energy balance satisfaction suggest the main qualitative data conclusions are acceptable.

5.7. Implications for Design and Applications

The results offer a practical guidance for the design of compact shell-and-coil heat exchangers operating with air over water coils. Firstly, the data confirms that the flow configuration and air-side velocity are strong systematic levers: counter-current operation and higher air speeds consistently increase the overall heat transfer coefficient and should be prioritised in design where fan power and piping layout allow. Secondly, geometry should be selected with the expected operating regime in mind rather than simply minimising pitch; the 24.5 mm coil delivered the highest U under steady high velocity, high flowrate conditions, whereas the 19 mm and 30 mm coils performed better at high velocity and lower flowrate, relative to other fluid conditions at these pitches.

Increasing water flowrate is not an efficient strategy in short coils, as demonstrated by this rig, because it reduces the water temperature drop and the driving force, often lowering U disregarding the higher waterside Reynolds number. Finally, while pulsed airflow has shown potential in the researched literature, the present experiment demonstrates that, in an external flow shell-and-coil configuration, it requires careful tuning of frequency and wave form to avoid performance deterioration. For most practical applications using a similar geometry and scale, a steady high-velocity airflow and a calculated pitch is likely to deliver the most reliable and efficient performance.

6. Conclusions and Future Work

6.1. Conclusions

This study experimentally investigated the effects of three coil pitches (19 mm, 24.5 mm and 30 mm), two water flowrates (1 and 2 L/min), two air velocities (10 and 18.5 m/s), two flow configurations (counter-current and parallel), and steady and pulsed airflow on the overall heat transfer coefficient of a helically coiled shell-and-coil heat exchanger. The following conclusions were drawn:

1. Counter-current flow outperformed parallel flow under all steady test conditions. The average improvement in U was 14.7% ranging from 3.4% to 30.2% depending on operating conditions.
2. Air velocity was the most influential single variable. Increasing air velocity from 10 m/s to 18.5 m/s raised U by circa 60% across almost all conditions. This is consistent with the air-side thermal resistance dominating the system: α_{air} was approximately an order of magnitude lower than α_{w} across all test conditions.

3. Increasing water flowrate from 1 to 2 L/min generally reduced U . The reduction in water temperature drops and LMTD at higher flowrate outweighed the marginal gain in waterside convection, since U is governed primarily by the airside resistance.
4. No single coil pitch was systematically optimal. The 24.5 mm pitch produced the highest measured U in this study ($0.245 \text{ kW/m}^2\text{K}$) at 18.5 m/s and 2 L/min air velocity and water flowrate, respectively. The 19 mm pitch performed well at high velocity and low flowrate; the 30 mm pitch was generally weaker but more sensitive to operating condition improvements. Each coil behaves as its own distinct heat exchanger whose relative performance depends on the operating point.
5. The selected Nusselt number correlations $Nu_{w,4}$ for the water side and $Nu_{air,2}$ showed accuracy with respect to the experimental U value, typically within 1 to 10%. The airside heat transfer coefficient was an order of magnitude lower than the water side, confirming the airside dominance of the thermal resistance.
6. Pulsed airflow at $St=0.2$ generally deteriorated the heat transfer performance. Enhancement ratios above 1 were observed only at 19 mm and 30 mm pitch with high water flowrate and low air velocity (+22.4% and +1.8%, respectively). The chosen oscillation parameters require further optimisation to achieve reliable and systematic enhancement.

6.2. Future Work

This study has demonstrated that coil pitch, flow configuration, air velocity, water flowrate and airflow conditions all influence the performance of a compact shell-and-coil heat exchanger. However, it has also highlighted several areas where further work is required to strengthen and generalise the findings. Firstly, only three coil pitches were tested, so future studies should investigate a wider and more finely spaced range of pitches, potentially including continuously varied coils to identify a true geometric optimal value rather than inferring it from a set constraint. Extending the experiments to different coil diameters and tube diameters would also help to separate the roles of pitch ratio and curvature ratio and to validate the applicability of the trends beyond the specific geometry of this rig.

The current analysis focused on thermal performance and did not measure pressure drop of the water. Future work should include pressure measurements so the thermo-hydraulic performance can be assessed using combined metrics such as heat transfer per unit pumping power. This would allow the optimisation of the water flowrates to be based on both heat transfer, and energy consumption which is crucial in real world applications. Additionally, more precise control of pulsed airflow is required – using a flow control device or variable speed fan that can generate well-defined square wave frequency profiles would allow the effect of frequency and waveform to be systematically mapped and compared directly with existing pulsating flow literature.

7. Risk Factor Summary

The key risks identified at the start of this project in the interim report were burns and scalds from hot water at approximately 60°C , electric shock from mains powered equipment within the vicinity of water, pipework under pressure and inconsistent coil pitch causing potentially non-uniform airflow resulting in unreliable data.

These risks were mitigated effectively throughout. Hot water connections were handled with insulated gloves, the system was always drained and depressurised before handling, and electrical equipment was kept out of the potential range of water. All connections were leak tested before experiments started and no leaking occurred during the experiments.

The pitch consistency risk proved to be the most significant. Early design iterations of the snap-fit inserts were not sufficient – allowing for a number of coil turn connections to disconnect, furthermore, failing

under load. This was identified during early design testing, and the formal risk assessment was reviewed and remained valid throughout testing with no new significant hazards arising during experimentation.

8. Reflective Summary

8.1. Self-Reflection

This project was a great experience in pushing my ability beyond content covered in lectures. I had to design a system that would work and have potential to be applicable or scaled in a real-world application – sourcing materials, iterating the insert design and conducting experimental tests to produce viable results. Having the ability to conduct experiments with a self-designed rig was something I am comfortable with, however, meticulous data recording was essential, as the reliability of all results depended on measurement accuracy. Where my assumptions going into experiments was backed up by theory and literature, to be contradicted in some cases by experimental data was interesting. Having to critically analyse the data and shifting from ‘confirming theory’ to scrutinising data is the more significant thing I have taken from conducting this project and something I can apply in potential future professional engineering practices.

8.2. Sustainability

Sustainability was considered throughout the project. The shell housing was a repurposed aluminium casing that would have otherwise been discarded, so too was the copper coil reducing material cost and consumption. The snap-fit inserts were 3D-printed from PLA, a recyclable biodegradable polymer and the design was iterated utilising CAD and prototypes, minimising waste. From an energy point of view, identifying the optimal operating conditions for this exchanger – specifically the pitch and fluid configuration that deliver the highest U per unit of airflow – directly supports the design of a more energy efficient heating and cooling system, which will reduce operational carbon emissions in real world applications.

8.3. Ethical Considerations

The main ethics this project relates to is safety and data integrity. Working with hot water at approximately 60°C, mains electrical equipment, and pressurised fluid in a laboratory environment legally required a formal risk assessment, which was completed and reviewed. Electrical connections were kept separately from water lines to minimise electric shocks and short-circuiting risks. I stored the results that minorly did not align with theoretical expectations, such as the U against pitch graphing but to thoroughly scrutinise them to find explanations or confirm it was not an anomaly. This reflects my professional engineering responsibility to be transparent in findings – especially when they may potentially compromise data processing and conclusions.

8.4. Equality, Diversity and Inclusion

While this project focuses on a physical engineering system opposed to a consumable product, the outcomes are applicable in high scale industries. Compact shell-and-coil heat exchangers are used in low-cost heating systems, industrial recovery process and domestic hot water systems frequently. By contributing to the understanding of how to improve their performance without major additional cost or complexity – through geometrical and fluid selection – this work has relevance to communities with limited infrastructure and budget.

9. Acknowledgements

I would like to thank the Heriot-Watt technical staff for their efforts in sourcing and providing the components and environment required to complete rig manufacture and conduct experiments over the

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Table 2 Correlations

Equations 6 to 9: Purandare, P.S., Lele, M.M. and Gupta, R. (2012). Parametric Analysis of Helical Coil Heat Exchanger. *International Journal of Engineering Research & Technology (IJERT)*, [online] 1(8), pp. 1-5.

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Table 3 Correlations

Equation 10: Salimpour, M.R. (2008) 'Heat transfer coefficients of shell and coiled tube heat exchangers', *Experimental Thermal and Fluid Science*, 33(2), pp. 203-207. Doi: [Heat transfer coefficients of shell and coiled tube heat exchangers](#) (Accessed: 3rd Apr. 2026)

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11. Appendices

Risk Assessment:

School/Institute/Directorate:	Location:	Reference/Title:	Date:	Review Date:
EPS	Riccarton, Edinburgh	Year 4 individual project, wind tunnel testing (testing forced convection)	20/01/2026	

Risk Rating Matrix (RR)	Likelihood(L)		
Severity (S)	Certain or near certain to occur (High)	Reasonably likely to occur (Medium)	Unlikely to Occur (Low)
Fatality, major injury or illness causing long term disability (High)	HIGH(H)	HIGH(H)	MEDIUM(M)
Injury or illness causing short term disability (Medium)	HIGH(H)	MEDIUM(M)	LOW(L)
Other Injury or Illness (Low)	MEDIUM(M)	LOW(L)	LOW(L)

Description:
Wind tunnel testing of an experimental rig using hot water and a heat exchanger.

What are the Hazards?	Who might be harmed?	Uncontrolled Risk Rating	Control measures (What are you already doing?)	Controlled Risk Rating	Responsible Person(s)
Hair or loose clothing getting caught in any moving parts/airflow	Operator	High	Long hair is tied back, no loose clothing is worn in the lab, and lab coats are always worn over normal clothes in the lab.	Low	Individual Operator
Steel rig or coil falling on feet	Operator	High	Minimise manual handling where gloves are used when doing so and safety boots with substantial toe protection (for example steel toe cap boots) are always worn in the workshop.	Low	Individual Operator
Any water/debris getting into eyes	Operator	Medium	Always wear safety glasses in the lab.	Low	Individual Operator
Ear damage from wind tunnel	Operator	Medium	Wear hearing plugs to protect ears whenever wind tunnel is turned on.	Low	Individual Operator
Scalding	Operator	High	Pipes should not be touched for any reason whilst boiling/near boiling water is flowing	Low	Individual Operator
Pinch points with pitch inserts	Operator	medium	Gloves are used when adding/removing inserts	Low	Individual Operator
Pressure changes in pipe – spillage & scald	Operator	High	Coil is drained prior to removal of inlet/outlet tubes and precaution is taken when removing tubes into bucket	Low	Individual Operator
Electricity in the vicinity of water – electric shock risk	Operator and/or equipment	High	Electrical equipment is elevated and moved beyond risk range of spraying water	Low	Individual Operator

Name		Position	Email	Signature
Author	Mackenzie Fettes	Student	Mf2036@hw.ac.uk	Mackenzie Fettes

Declaration

I confirm that I have read the Risk Assessment, I understand the hazards, and risks involved and will follow all the control measures stated. If I have any queries or concerns, I will contact the above.

Name	Organisation	Staff/Student/Visitor	Signature	Date
Mackenzie Fettes	HWU	Student	Mackenzie Fettes	20/01/2026